

Controllers and Final Control Elements

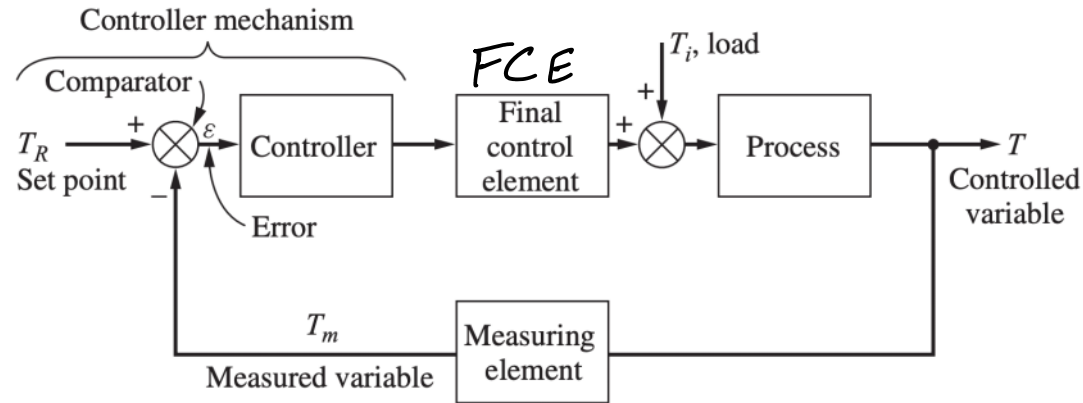


FIGURE 8-2
Block diagram of a simple control system.

Controller input is error
output is signal for FCE
e.g air pressure
to be applied on
a diaphragm in valve

Pneumatic control valve
with a
Proportional controller

Mechanisms

→ Control Valve

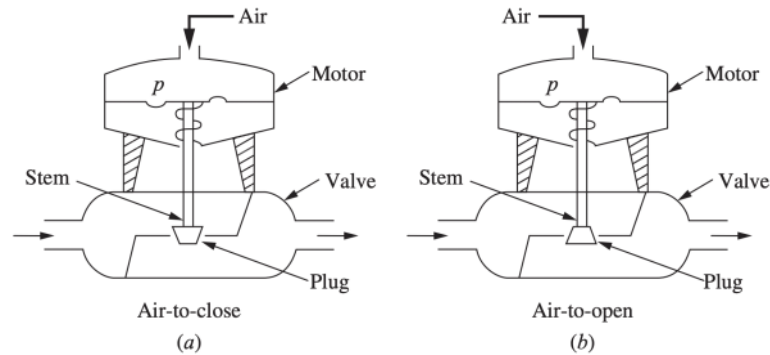


FIGURE 9-1
Pneumatic control valves. (a) Air to close; (b) air to open.

- ↪ Choice based on safety
 - Fail open for cooling applications
- ↪ Stem position is proportional to pressure applied
- ↪ Linear valve
 - flow proportional to valve top pneumatic pressure

→ Controller

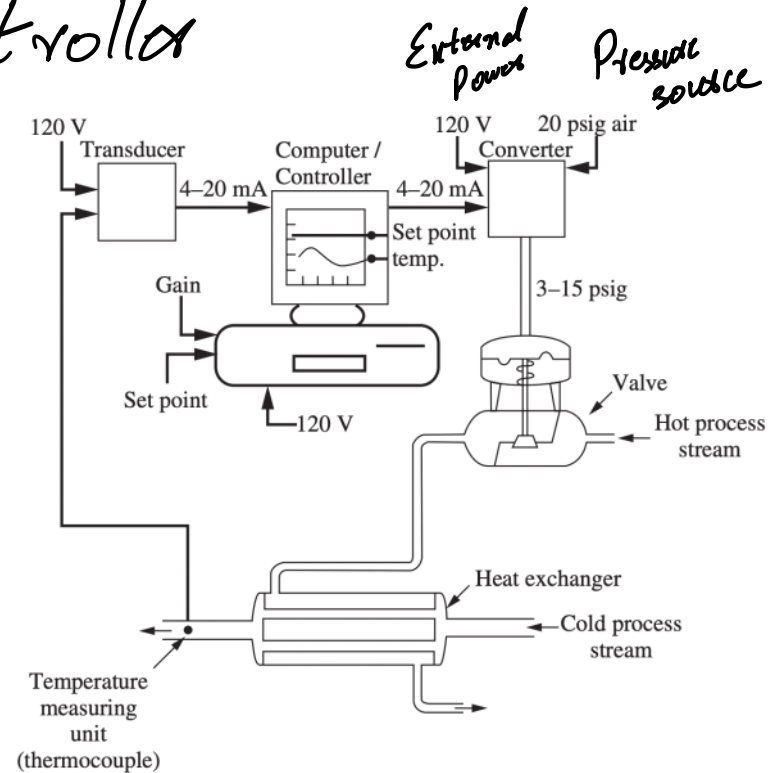


FIGURE 9-2
Schematic diagram of control system.

- Transducer → Temp to current
- Computer / Controller → Current to Current
- Converter → Current to Pressure
- Control valve → Pressure to flow rate

Reduction in input Temp of Cold stream

→ Thermocouple detects a decrease in outlet temp and produces proportional change in signal to controller

→ Controller output increases proportionally sent to converter

→ Output pressure from converter increases and flow rate increases

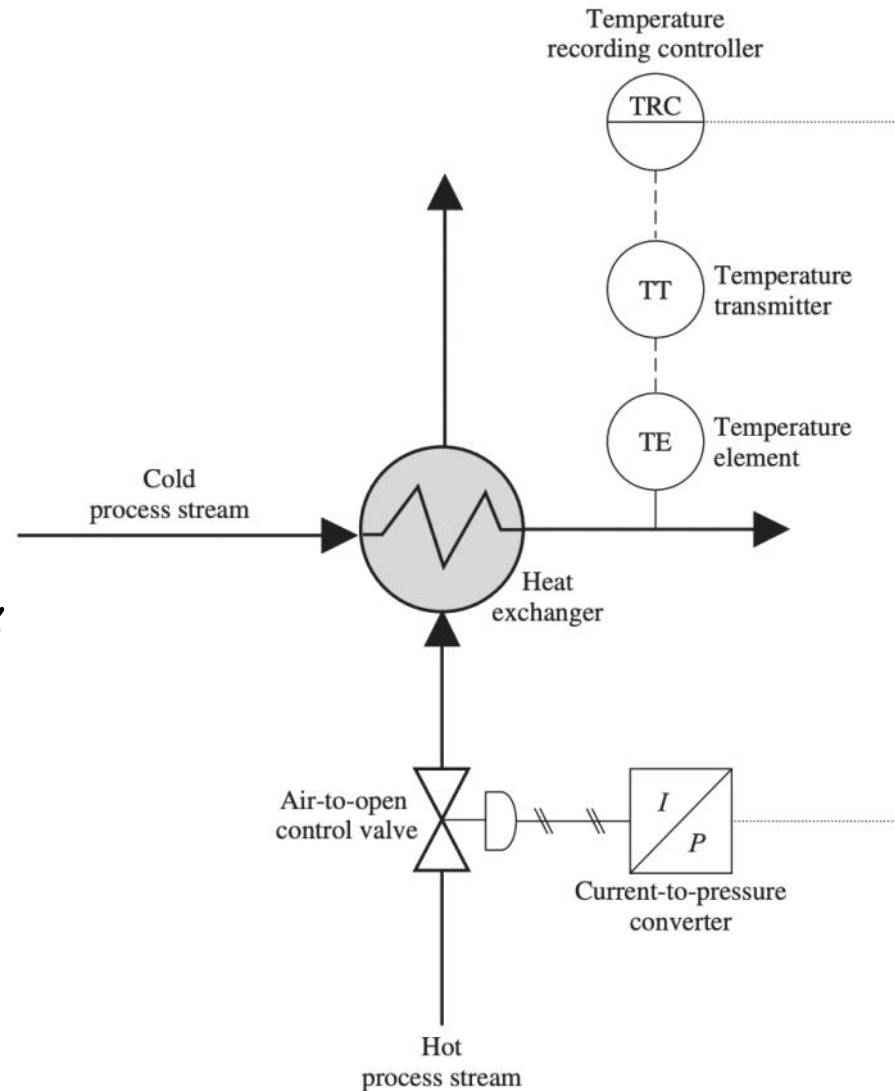


FIGURE 9-3

Piping and instrumentation diagram for control system of Fig. 9-2.

Aim is to return output temp to desired temperature

For simplification; lumping
Transducer }
Controller } — Controller
Converter }

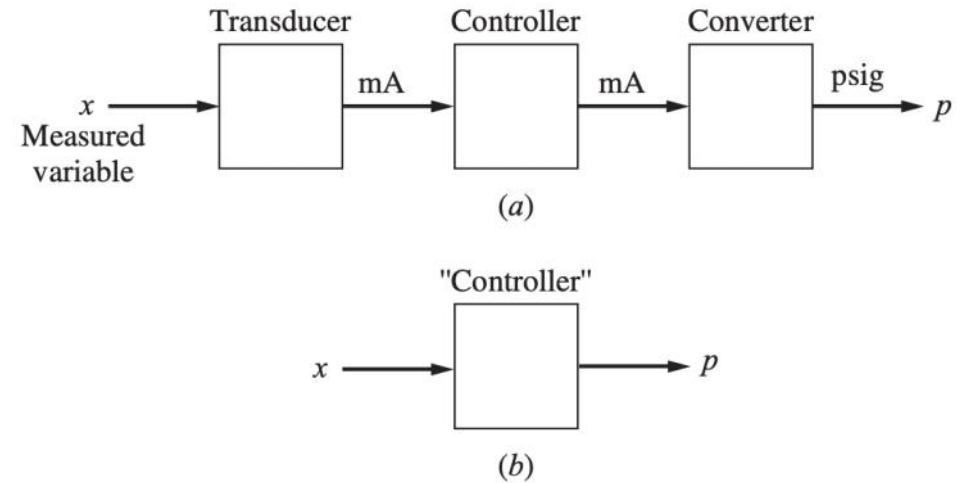


FIGURE 9-4
Equivalent block for transducer, controller, and converter.

Ideal Transfer Functions

→ Control Valve

↳ First order transfer function

$$\frac{Q(s)}{P(s)} = \frac{K_v}{\tau_v s + 1}$$

$K_v \rightarrow$ Gain

→ Const of proportionality between
l.s flow rate and valve-top pressure

$\tau_v \rightarrow$ Time constant of valve

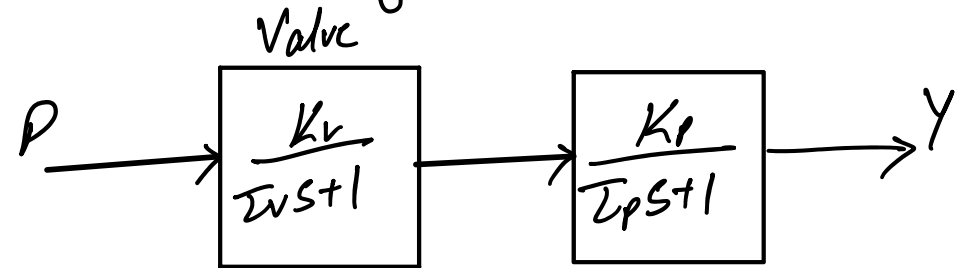
In Practical Systems $\tau_v \ll \tau_p$

$$\Rightarrow \frac{Q(s)}{P(s)} = K_v$$

Valve contributes negligible
dynamic lag

For first order valve and process connected in series

Block Diagram



Giving Transfer function

$$\frac{Y(s)}{P(s)} = \frac{K_v}{\tau_v s + 1} \frac{K_p}{\tau_p s + 1}$$

For unit step pressure change

$$Y(s) = \frac{1}{s} \frac{K_v K_p}{(\tau_v s + 1)(\tau_p s + 1)}$$

$$Y(s) = \frac{1}{s} \frac{K_v K_p}{(\tau_v s + 1)(\tau_p s + 1)}$$

Inverse

$$y(t) = K_v K_p \left[1 - \frac{\tau_v \tau_p}{\tau_v - \tau_p} \left(\frac{1}{\tau_p} e^{-t/\tau_v} - \frac{1}{\tau_v} e^{-t/\tau_p} \right) \right]$$

Considering $\tau_v \ll \tau_p$

$$y(t) = K_v K_p (1 - e^{-t/\tau_p})$$

Simple transfer function

$$\frac{Y(s)}{P(s)} = K_v \frac{K_p}{\tau_p s + 1}$$

Combination of Process and Valve is first order

→ Controllers

↳ lumped Transducer
Controller
Converter

↳ Input → Measured variable (x)

↳ Output → Pneumatic signal (P)

Proportional Control

↳ Simplest controller

↳ Goal is to eliminate error

→ Proportional control can
reduce error but can not
eliminate it

↳ Controller gain is adjustable
parameter

↳ output is a signal → pressure

$$P = K_c E + P_s$$

↓
Proportional Gain
OR
Sensitivity

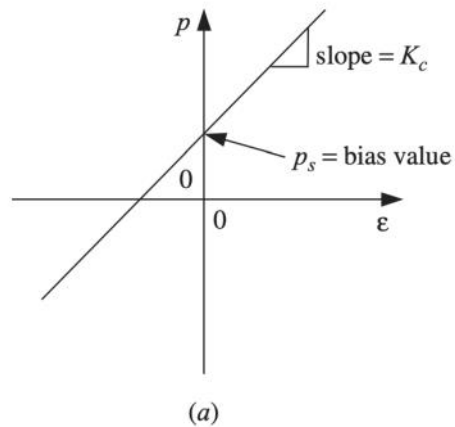
$$P - P_s = K_c E = P$$

Laplace

$$\boxed{\frac{P(s)}{E(s)} = K_c}$$

Proportional
Controller
transfer function

Proportional controller—Ideal behavior



Proportional controller—actual behavior

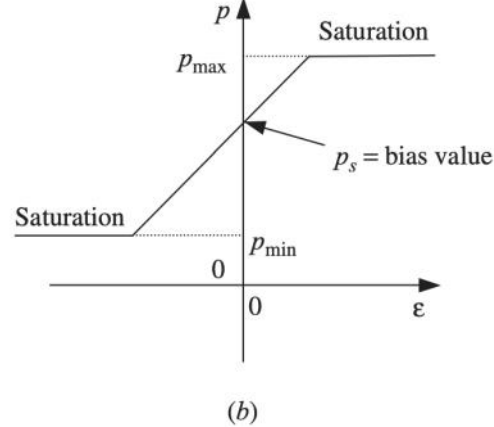


FIGURE 9-6

Proportional controller output as a function of error input to the controller. (a) Ideal behavior; (b) actual behavior.

Upper Saturation
15 psig or 20mA
Lower Saturation
3 psig or 4mA

Example 9.1. A pneumatic proportional controller is used in the process shown in Fig. 9-2 to control the cold stream outlet temperature within the range of 60 to 120°F. The controller gain is adjusted so that the output pressure goes from 3 psig (valve fully closed) to 15 psig (valve fully open) as the measured temperature goes from 71 to 75°F with the set point held constant. Find the controller gain K_c .

Starting with

$$K_c = \frac{P(\text{psi})}{E(\text{°F})}$$

$$\text{Gain} = \frac{\Delta P}{\Delta E} = \frac{15 - 3 \text{ psi}}{75 - 71 \text{ °F}} = 3 \text{ psi/°F}$$

If gain changed to 0.4 psi/°F
What is error that will cause control valve to go from fully open to close

$$\Delta T = \frac{\Delta P}{\text{Gain}} = \frac{15 - 3 \text{ psi}}{0.4 \text{ psi/°F}} = 30 \text{ °F}$$

Valve fully opens if error reaches 30°F

ON/OFF Control

→ Rapid ON/OFF "chattering"
→ Avoided by introduction of
"Dead band"

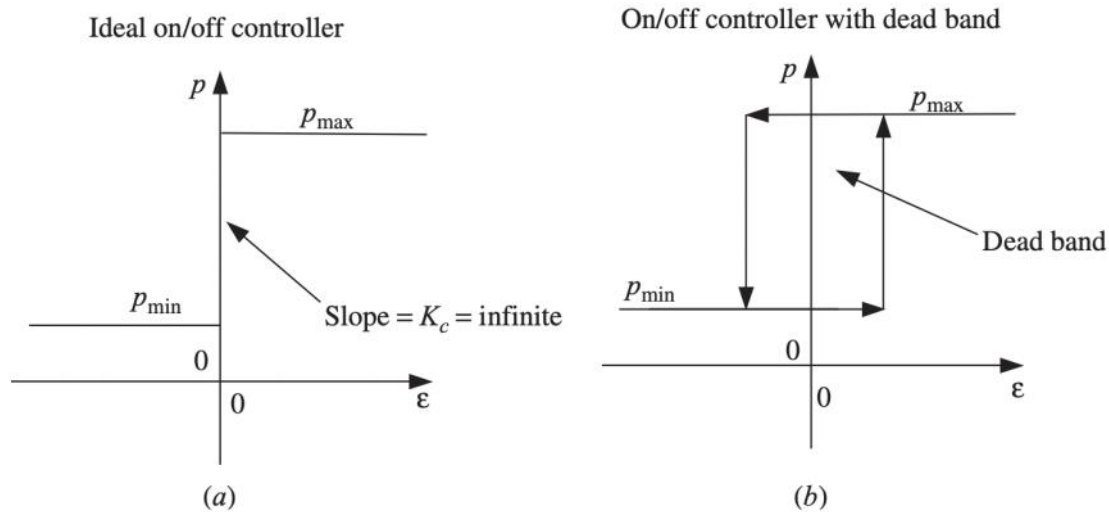


FIGURE 9-7

Output from on/off controller as a function of error input to the controller. (a) Ideal on/off controller; (b) on/off controller with dead band.

- Special case of proportional control
- For very high Gain → ON/OFF
- Valve acts as a switch
- With "dead band" → Error must reach some value for action

Proportional Integral (PI) Control

→ Proportional control only results in residual error

→ Integral mode ultimately drives the error to zero → Integral time can be adjusted

$$P = K_c \varepsilon + \frac{K_c}{T_I} \int_0^t \varepsilon dt + P_s$$

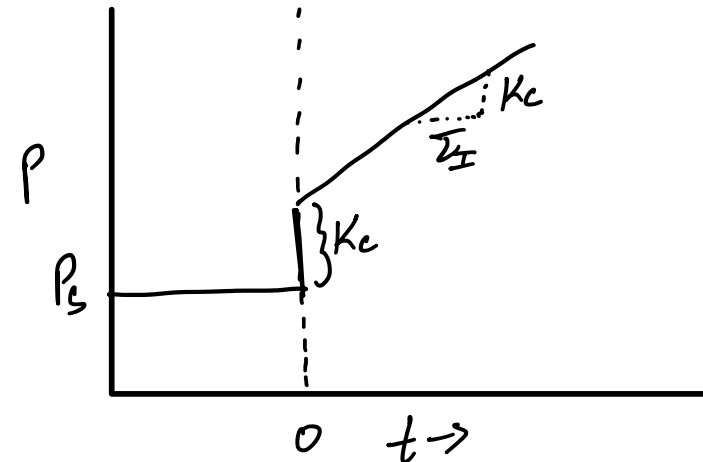
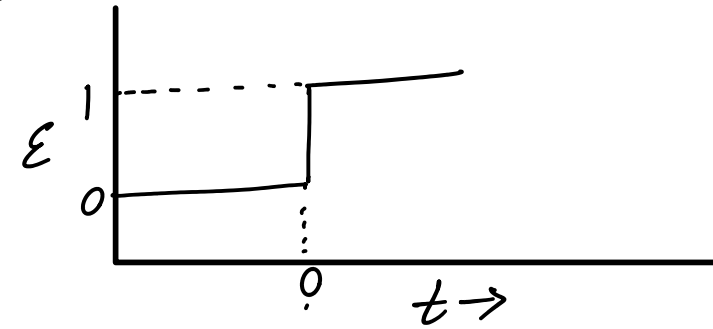
Proportional gain

Integral time

term proportional to integral of error

Bias value

For $\varepsilon = 1$



$$\begin{aligned} \text{Proportional term} &= K_c \varepsilon = K_c \\ \text{Integral term} &= \frac{K_c}{T_I} \int_0^t \varepsilon dt = \frac{K_c}{T_I} t \\ \Rightarrow P(t) &= K_c + \frac{K_c}{T_I} t + P_s \end{aligned}$$

P changes suddenly by amount K_c and then changes linearly with time at rate $\frac{K_c}{\tau_I}$

Transfer function for:-

$$P = K_c E + \frac{K_c}{\tau_I} \int_0^t E dt + P_s$$

$$P - P_s = K_c E + \frac{K_c}{\tau_I} \int_0^t E dt$$

$$P = K_c \left(E + \frac{1}{\tau_I} \int_0^t E dt \right)$$

Laplace

$$P(s) = K_c \left(E(s) + \frac{1}{\tau_I} \frac{E(s)}{s} \right)$$

$$\mathcal{L} \left\{ \int_0^t f(\tau) d\tau \right\} = \frac{F(s)}{s}$$

$$P(s) = K_c E(s) \left(1 + \frac{1}{\tau_I s} \right)$$

$$\frac{P(s)}{E(s)} = K_c \left(1 + \frac{1}{\tau_I s} \right)$$

Proportional Integral Controller
Transfer function

* $\frac{1}{\tau_I} \Rightarrow$ Reset rate

Proportional Derivative Control (PD)

→ Derivative control acts on derivative of error
→ Most active when error changes rapidly

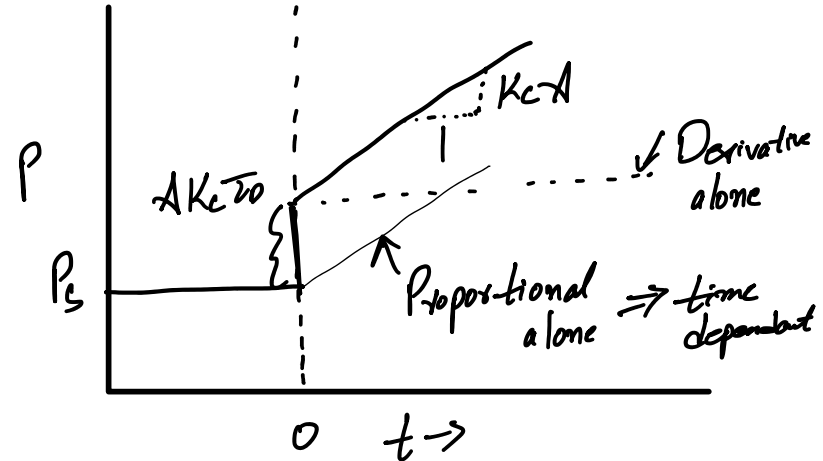
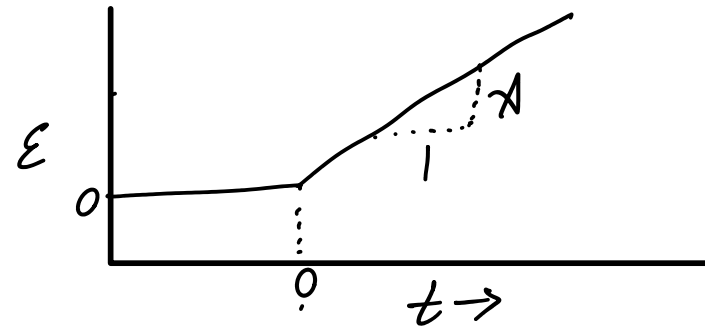
$$P = K_c \epsilon + K_c \tau_D \frac{d\epsilon}{dt} + P_s$$

Derivative
time

term proportional
to derivative of
error

→ Also called
→ Rate control
→ Anticipatory Control

For $\epsilon(t) = \lambda t$



$$P(t) = \lambda K_c t + \lambda K_c \tau_D + P_s$$

Transfer function

$$P = K_c E + K_c \tau_D \frac{dE}{dt} + P_s$$

$$P - P_s = K_c E + K_c \tau_D \frac{dE}{dt}$$

$$P = K_c \left(E + \tau_D \frac{dE}{dt} \right)$$

$$P(s) = K_c \left(E(s) + \tau_D s E(s) \right)$$

$$\frac{P(s)}{E(s)} = K_c (1 + \tau_D s)$$

Proportional derivative
controller transfer function

→ Derivative action is based on how rapidly
the error is changing

Proportional Integral Derivative (PID) Control

Combining all three

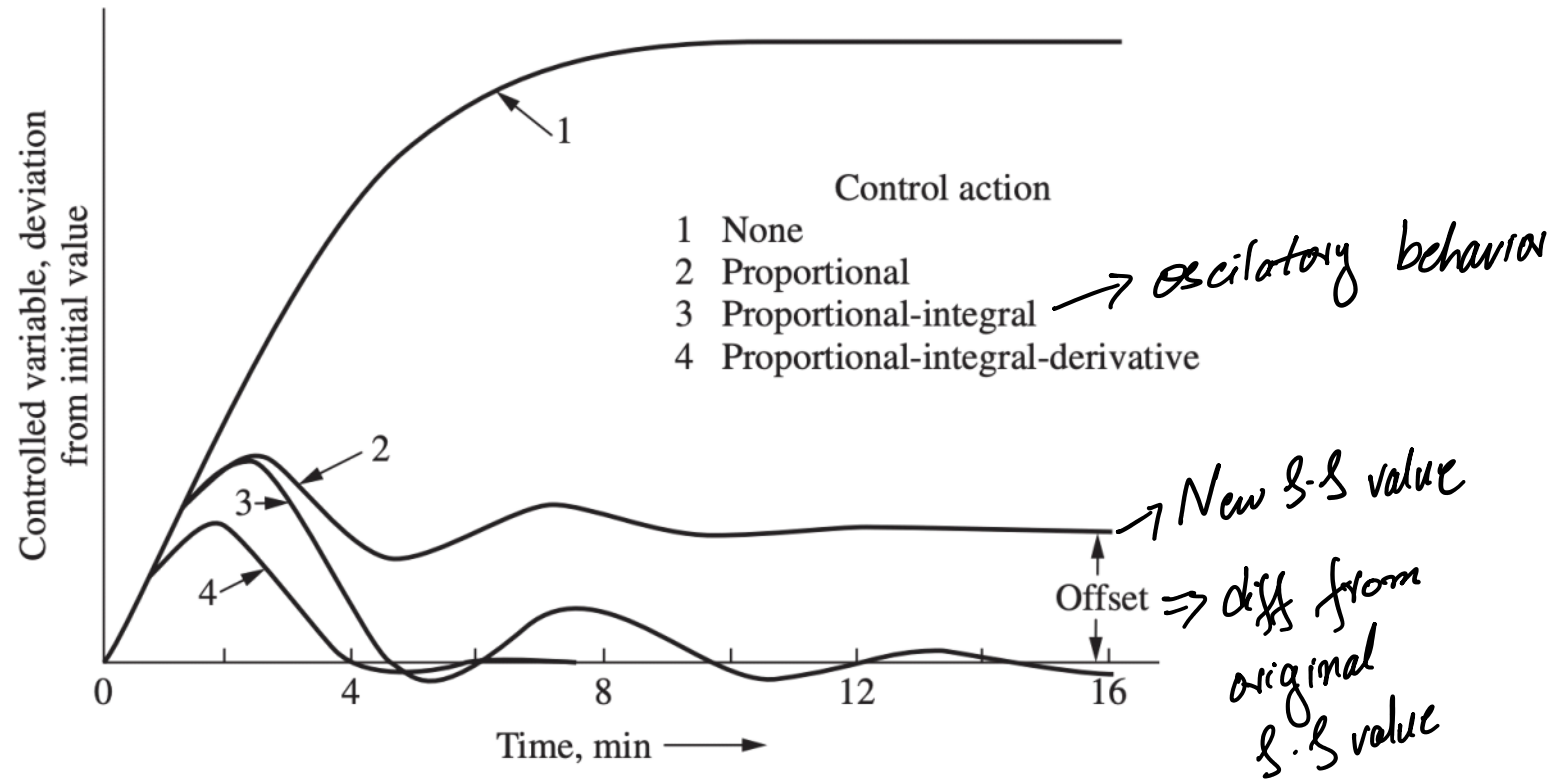
$$P = K_c E + \frac{K_c}{\tau_I} \int_0^t E dt + K_c \tau_D \frac{dE}{dt} + P_s$$

With transfer function

$$\frac{P(s)}{E(s)} = K_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right)$$

K_c , τ_I and τ_D can be adjusted

Response as a result of permanent disturbance



→ Proportional is acceptable if offset is acceptable

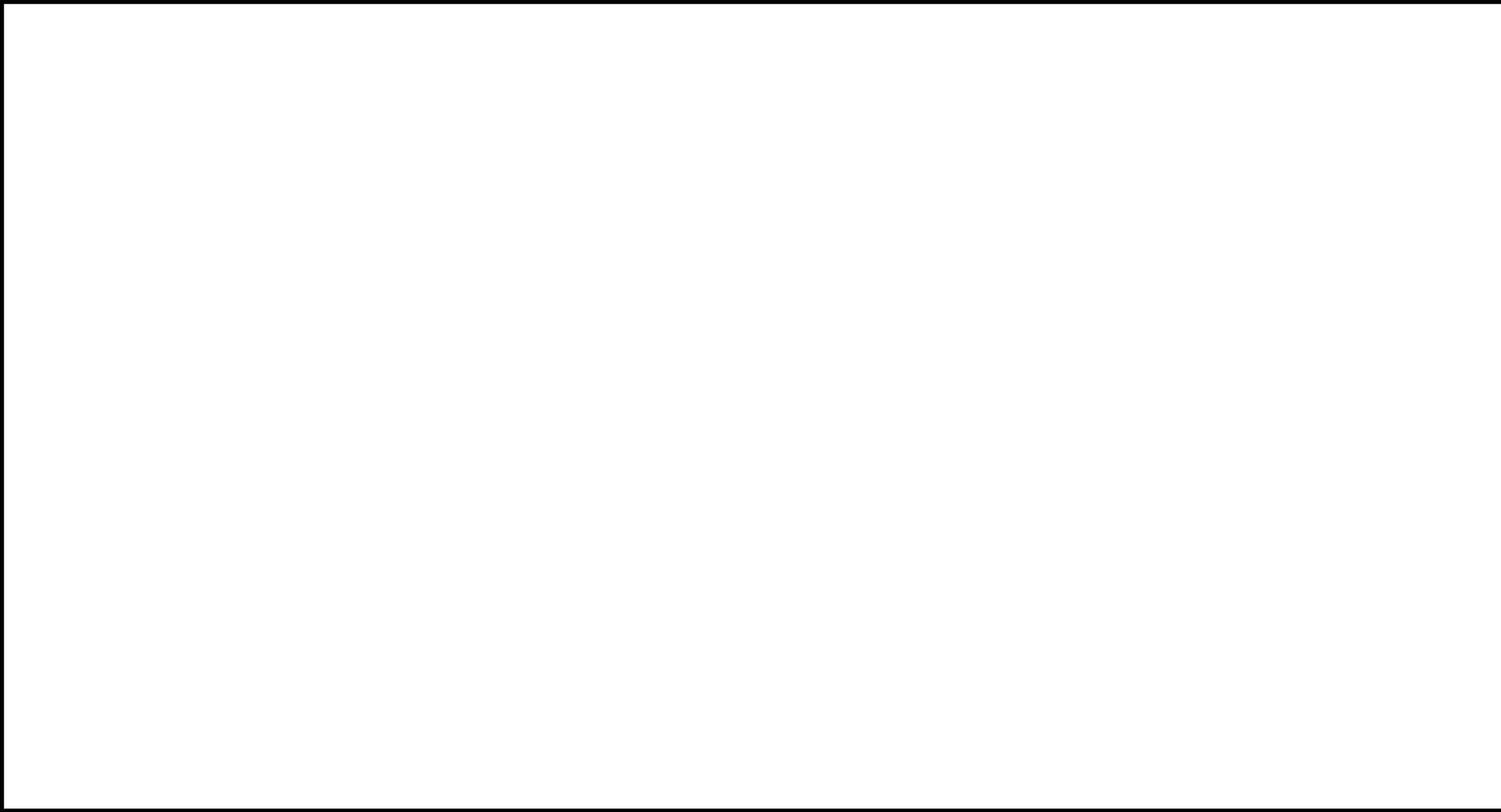
→ Integral action to get rid of offset

→ Derivative action added to remove oscillations

FIGURE 9-10

Response of a typical control system showing the effects of various modes of control.

Controller type	Time domain model	Transfer function
Proportional (P)	$p = K_c \varepsilon + p_s$	$\frac{P(s)}{\varepsilon(s)} = K_c$
Proportional-integral (PI)	$p = K_c \varepsilon + \frac{K_c}{\tau_I} \int_0^t \varepsilon dt + p_s$	$\frac{P(s)}{\varepsilon(s)} = K_c \left(1 + \frac{1}{\tau_I s} \right)$
Proportional-derivative (PD)	$p = K_c \varepsilon + K_c \tau_D \frac{d\varepsilon}{dt} + p_s$	$\frac{P(s)}{\varepsilon(s)} = K_c (1 + \tau_D s)$
Proportional-integral-derivative (PID)	$p = K_c \varepsilon + K_c \tau_D \frac{d\varepsilon}{dt} + \frac{K_c}{\tau_I} \int_0^t \varepsilon dt + p_s$	$\frac{P(s)}{\varepsilon(s)} = K_c \left(1 + \tau_D s + \frac{1}{\tau_I s} \right)$



Stirred Tank
Heater

Closed Loop Transfer Functions

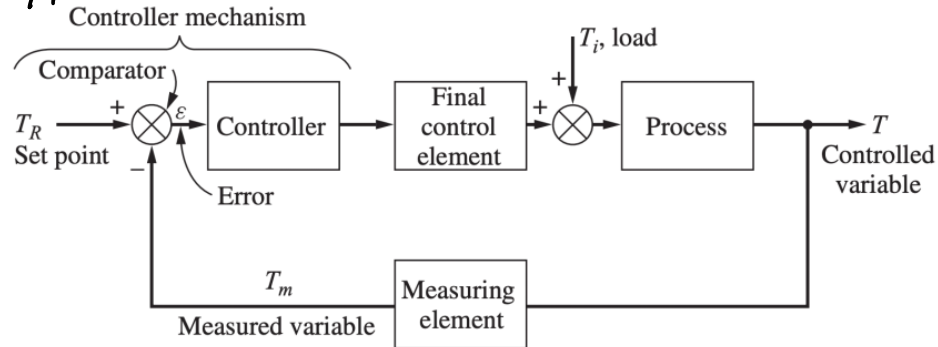


FIGURE 8-2
Block diagram of a simple control system.

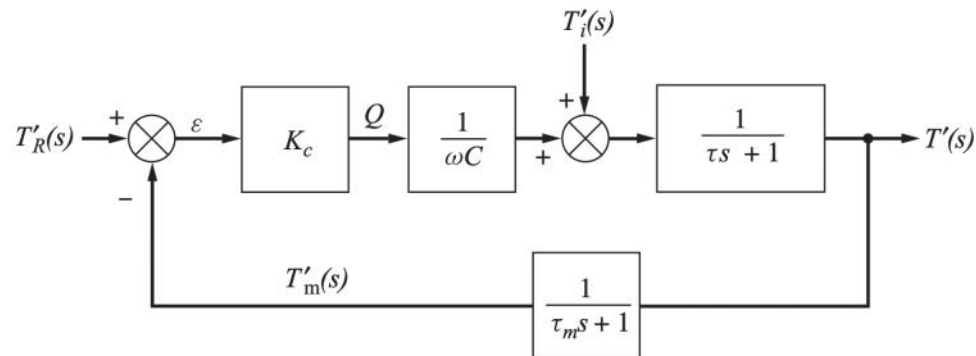


FIGURE 8-16
Block diagram of control system.

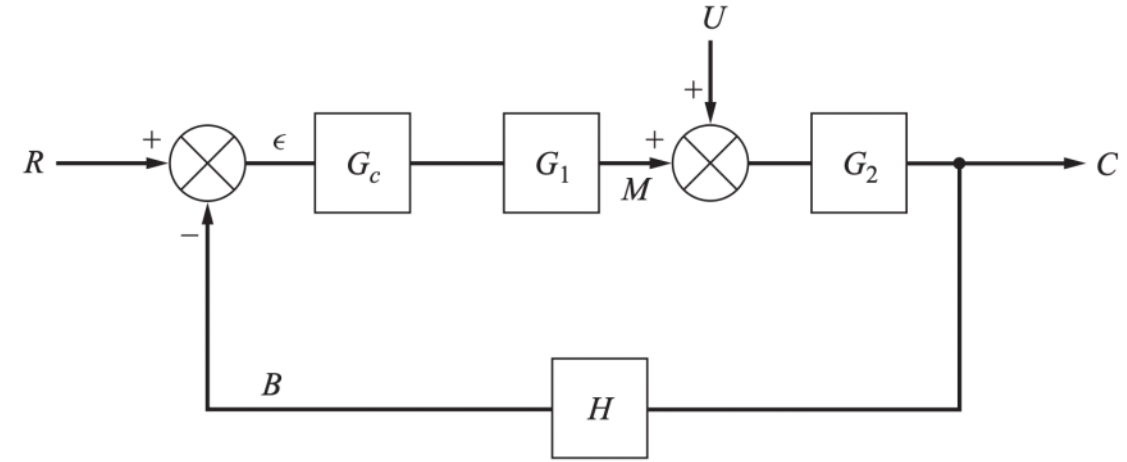


FIGURE 11-1
Standard control system nomenclature.

- R = set point or desired value
- C = controlled variable
- ϵ = error
- B = variable produced by measuring element
- M = manipulated variable
- U = load variable or disturbance
- G_c = transfer function of controller
- G_1 = transfer function of final control element
- G_2 = transfer function of process
- H = transfer function of measuring element

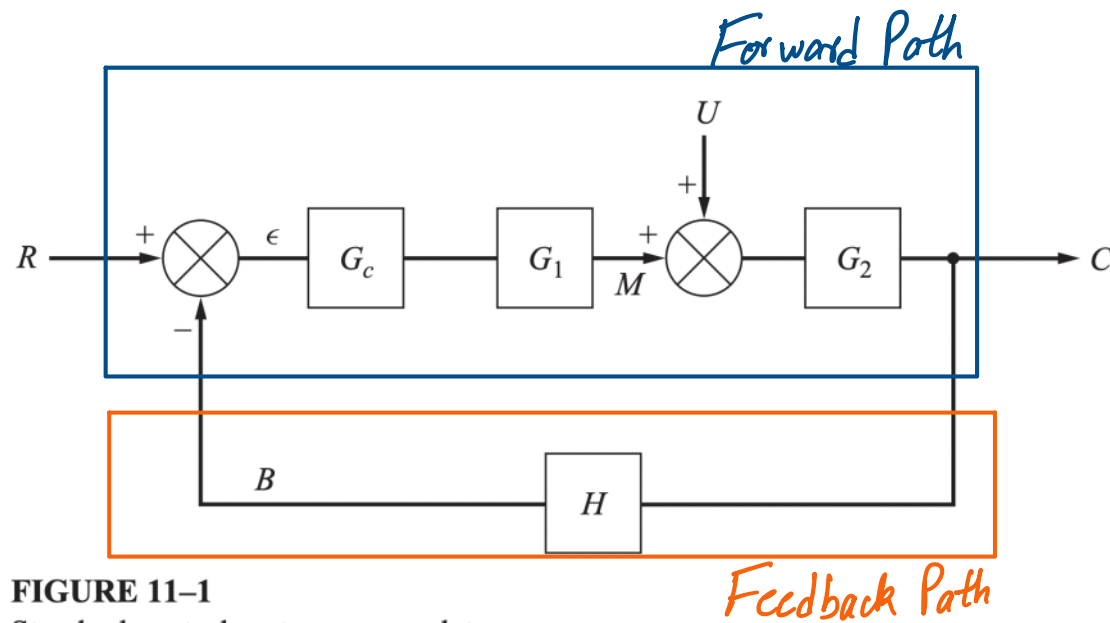
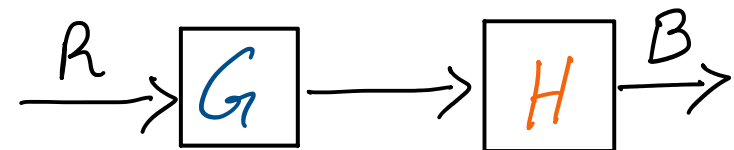


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 G_2 = transfer function of process
 H = transfer function of measuring element

G → used for transfer functions in forward path
 H → used for transfer function in feedback path

G, H → Open loop transfer function
 → Product of all transfer functions
 → Relates B to R if feedback loop is disconnected from comparator → open



→ We now focus on when the loop is closed

Multi loop system
 → Cascade Control → Multiple Loads
 → Several Feedback Paths

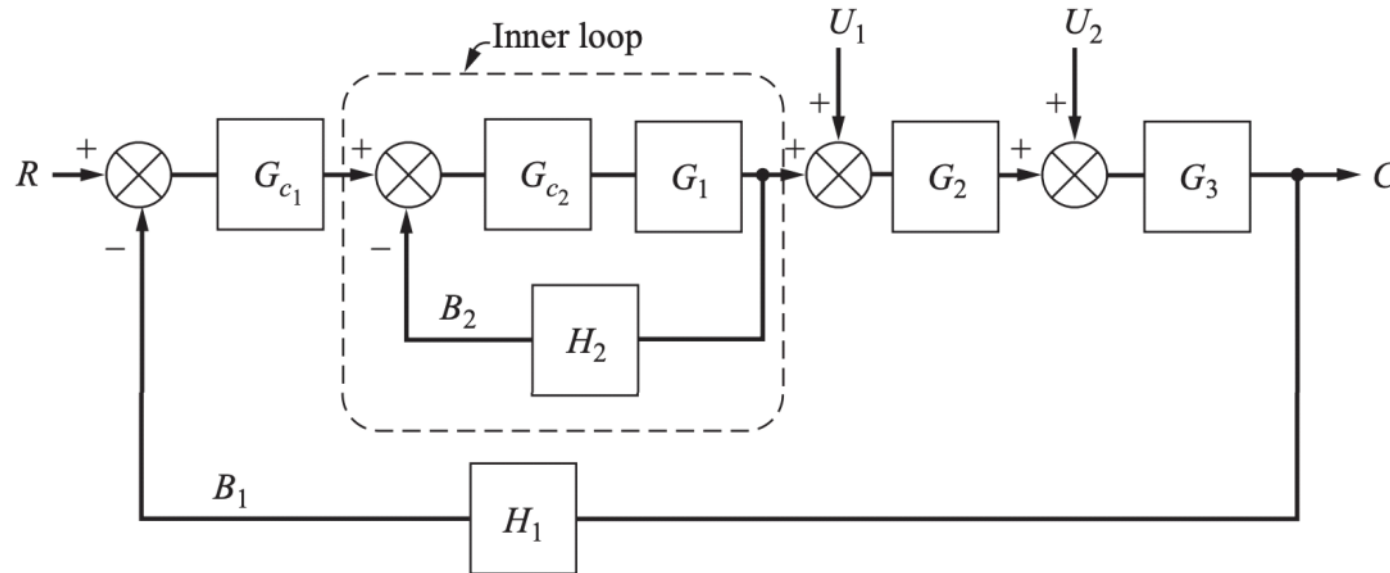


FIGURE 11-2
 Block diagram for a multiloop, multiload system.

Overall Transfer Function for Single-Loop Systems

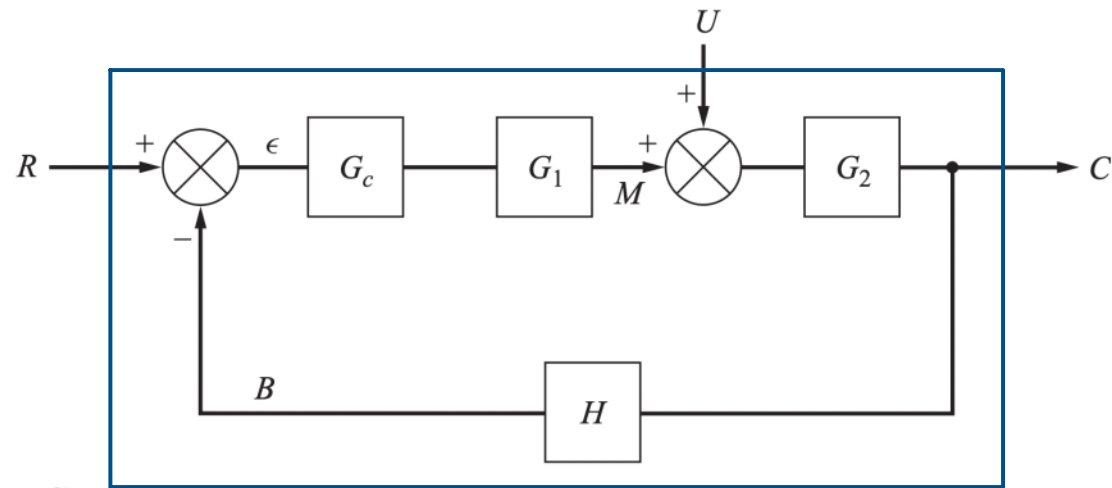


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- G_1 = transfer function of final control element
- G_2 = transfer function of process
- H = transfer function of measuring element

→ Transfer function Relating C to R
OR C to U

→ Determine Response of C to changes in R and U

→ Response to R @ $U=0$

- Servo Problem

→ Response to U @ $R=0$

- Regulator Problem

Overall Transfer function for change in Set Point ; $U=0$

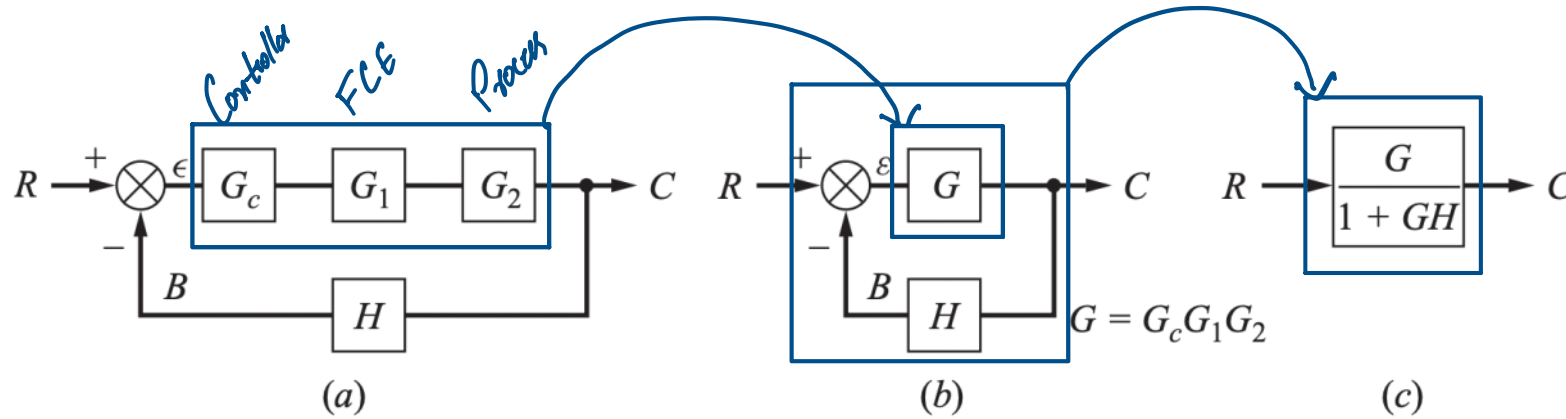
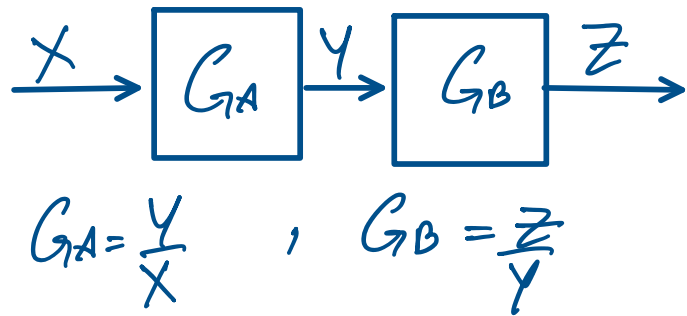


FIGURE 11-3
Block diagram reduction to obtain overall transfer function.

Multiplication of transfer func.



$$\frac{Z}{X} = G_A G_B$$

$$G = \frac{C}{E}, \quad H = \frac{B}{C} \Rightarrow B = HC$$

$$\text{where } E = R - B$$

$$C = G(R - B) = G(R - HC)$$

$$C = GR - GHC$$

$$C + GHC = GR$$

$$\boxed{\frac{C}{R} = \frac{G}{1+GH}}$$

Overall Transfer Function for change in Load ; $R=0$

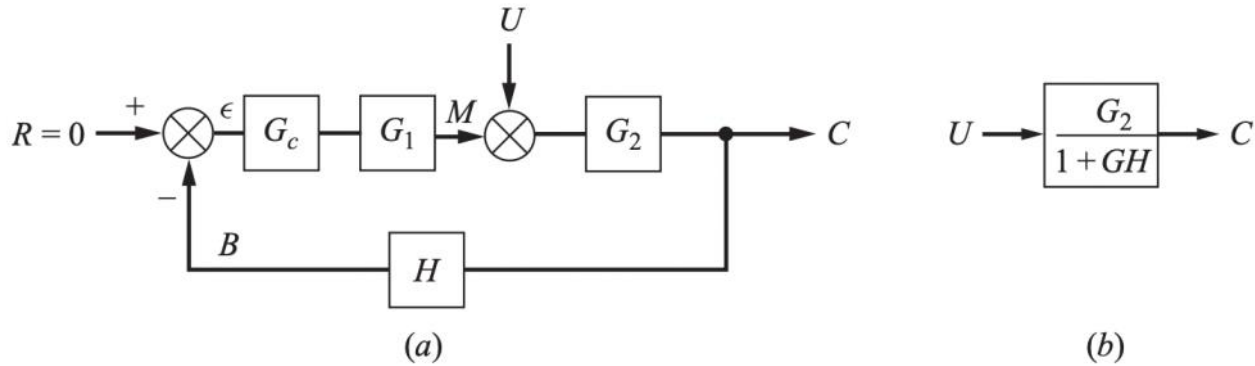


FIGURE 11-5

Block diagram for change in load.

$$G_2 = \frac{C}{U+M} \Rightarrow C = G_2(U+M)$$

$$G_c G_1 = \frac{M}{E} \Rightarrow M = G_c G_1 E$$

$$H = \frac{B}{C} \Rightarrow B = HC$$

$$E = R - B = -B$$

$$C = G_2(U + G_c G_1 E)$$

$$C = G_2(U + G_c G_1 (-HC))$$

$$C = G_2 U + G_c G_1 G_2 (-HC)$$

$$C = G_2 U - G_c G_1 G_2 HC$$

$$C + G_c G_1 G_2 HC = G_2 U$$

$$C(1 + G_c G_1 G_2 H) = G_2 U$$

$$\frac{C}{U} = \frac{G_2}{1 + G_c G_1 G_2 H}$$

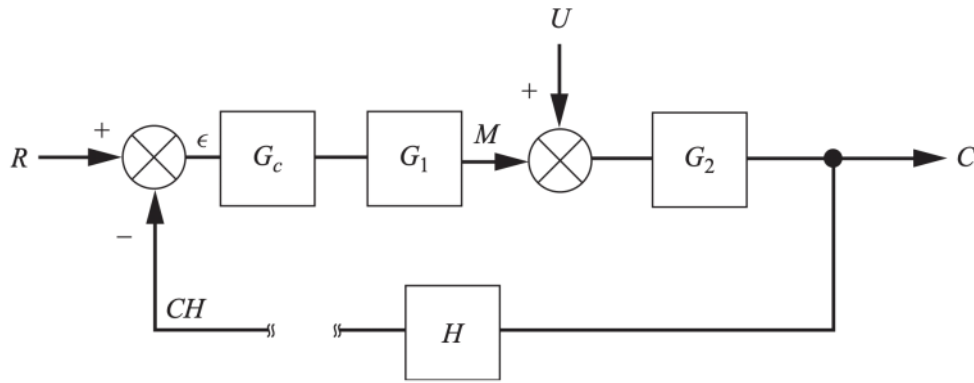


FIGURE 11-6
Modified block diagram for determining closed-loop transfer function.

$$E = R - CH$$

$$G_c = \frac{G_c (R - CH)}{E}$$

$$G_1 = \frac{M}{G_c (R - CH)} \Rightarrow M = G_1 G_c (R - CH)$$

$$G_2 = \frac{C}{M + U} \Rightarrow C = G_2 (M + U)$$

$$C = G_2 (M + U)$$

$$C = G_2 (G_1 G_c (R - CH) + U)$$

$$C = \underbrace{G_2 G_1 G_c}_{G_1} R - \underbrace{G_c G_1 G_2}_{G_1} CH + G_2 U$$

$$C + G_1 CH = G_1 R + G_2 U$$

$$C (1 + G_1 C) = G_1 R + G_2 U$$

$$C = \frac{G_1}{(1 + G_1 C)} R + \frac{G_2}{(1 + G_1 C)} U$$

Transfer function relating any X and Y is obtained by

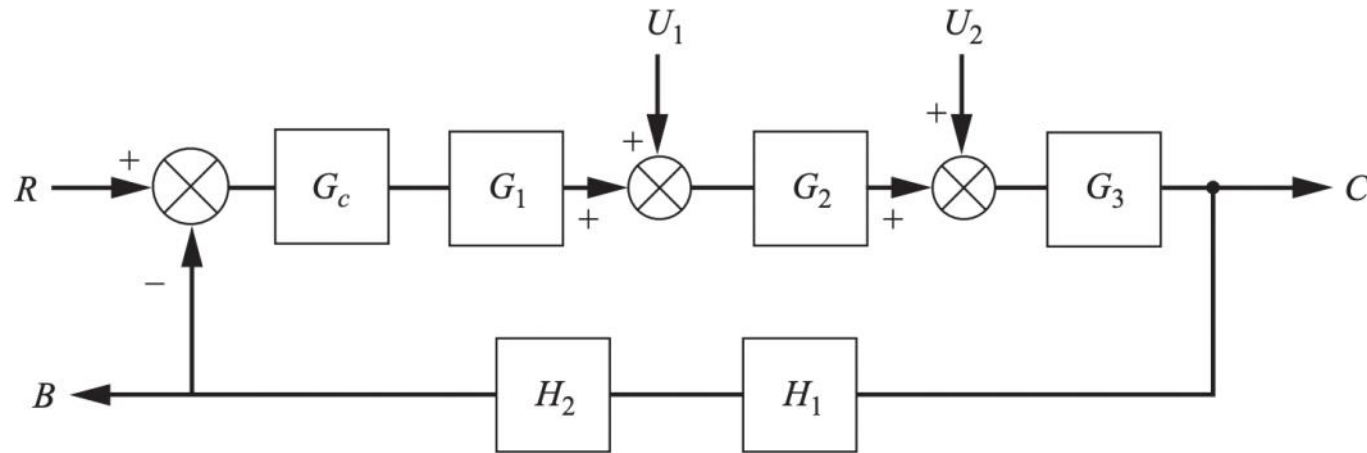
$$\frac{Y}{X} = \frac{\pi_{\text{forward}}}{1 + \pi_{\text{loop}}} \quad (\text{negative feedback})$$

$$\frac{Y}{X} = \frac{\pi_{\text{forward}}}{1 - \pi_{\text{loop}}} \quad (\text{positive feedback})$$

π_{forward} = Product of Transfer functions in forward path between locations of X and Y

π_{loop} = Product of all transfer functions in the loop

Example 11.1. Determine the transfer functions C/R , C/U_1 , and B/U_2 for the system shown in Fig. 11–7. Also determine an expression for C in terms of R and U_1 for the situation when both set point change and load change occur simultaneously.



$$\frac{C}{R} = \frac{\pi_{\text{fwd}}}{1 + \pi_{\text{loop}}} = \frac{G_c G_1 G_2 G_3}{1 + \underbrace{G_c G_1 G_2 G_3 H_2 H_1}_G} = \frac{G_c G_1 G_2 G_3}{1 + G}$$

$$\frac{C}{U_1} = \frac{G_2 G_3}{1 + G} \quad ; \quad \frac{B}{U_2} = \frac{G_3 H_1 H_2}{1 + G}$$

Expression of C in terms of R and U_1

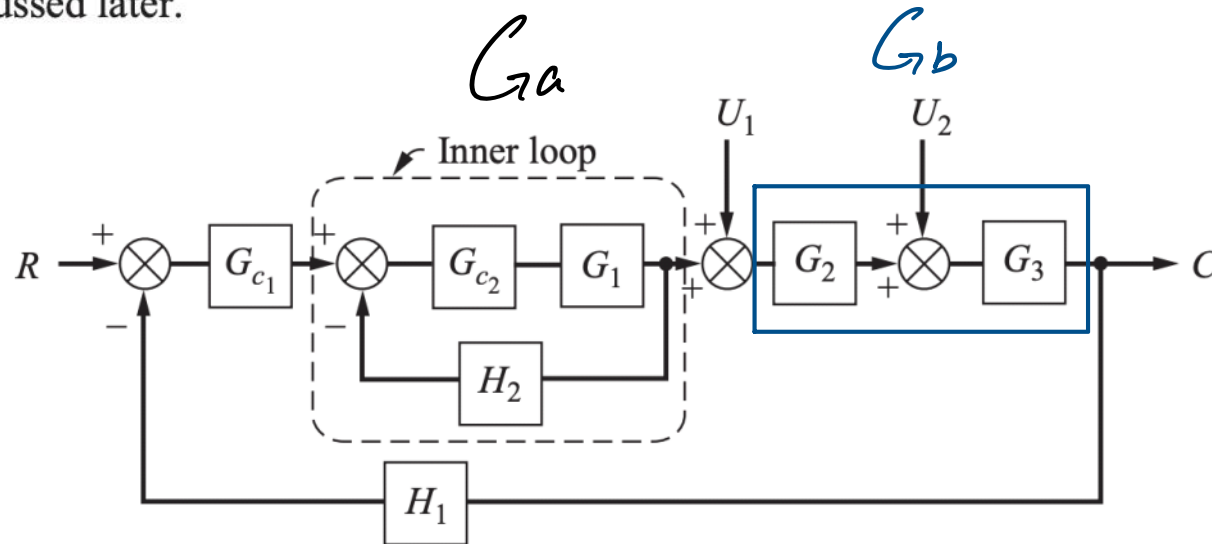
$$\frac{C}{R} = \frac{G_c G_1 G_2 G_3}{1+G}$$

$$\frac{C}{U_1} = \frac{G_2 G_3}{1+G}$$

$$C = \frac{G_c G_1 G_2 G_3}{1+G} R + \frac{G_2 G_3}{1+G} U_1$$

Overall Transfer functions for Multiloop Control Systems

Example 11.2. Determine the transfer function C/R for the system shown in Fig. 11-8. This block diagram represents a cascade control system, which will be discussed later.



Overall

$$\frac{C}{R} = \frac{G_{c1} G_a G_b}{1 + G_{c1} G_a G_b H_1}$$

For Inner loop first

$$G_a = \frac{G_{c2} G_1}{1 + G_{c2} G_1 H_2}$$

For Processes G_2 and G_3

$$G_b = G_2 G_3$$

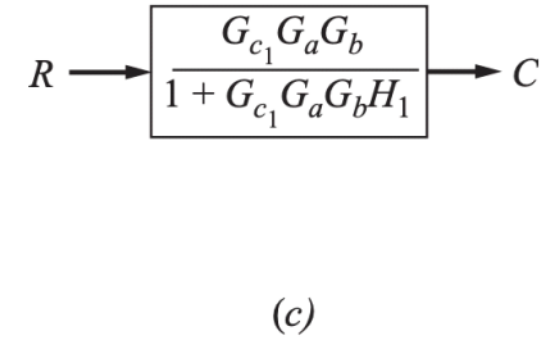
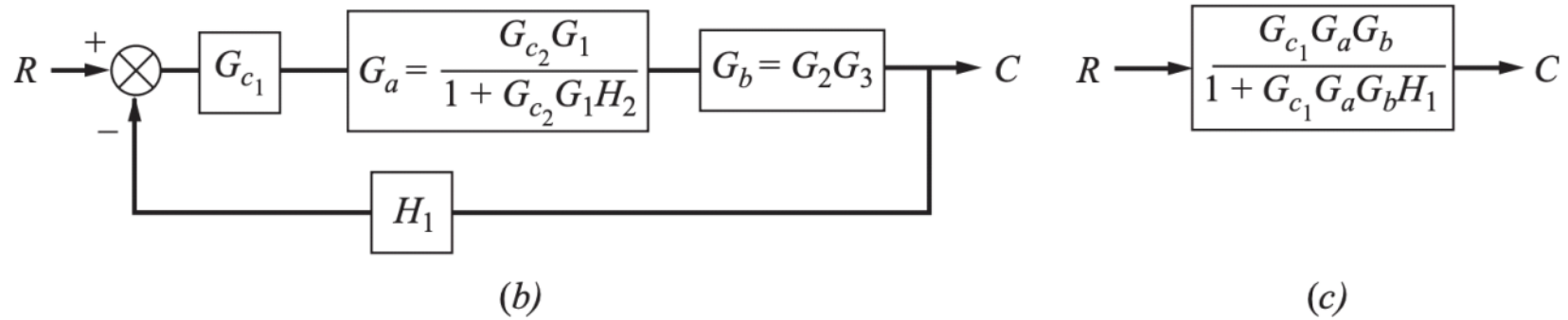
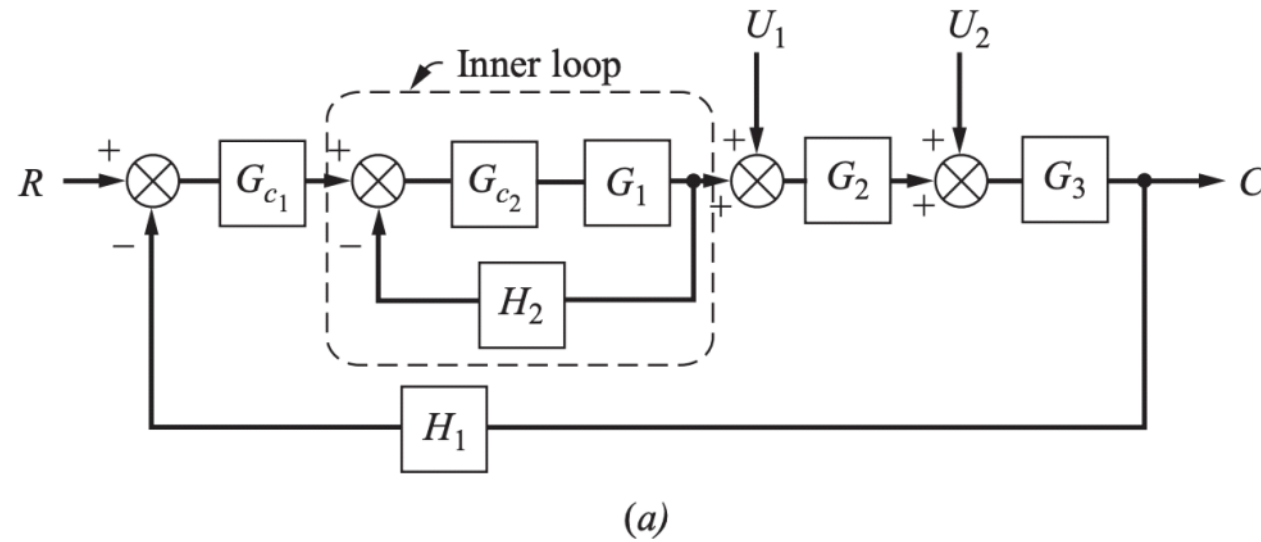


FIGURE 11–8

Block diagram reduction. (a) Original diagram; (b) first reduction; (c) final block diagram.